MEDIUM

PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

05

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS

At a movie theater, there are a total of 350 customers. Each customer is located in either theater A, theater B, or theater C. If one of these customers is selected at random, the probability of selecting a customer who is located in theater A is 0.48, and the probability of selecting a customer who is located in theater B is 0.24. How many customers are located in theater C?

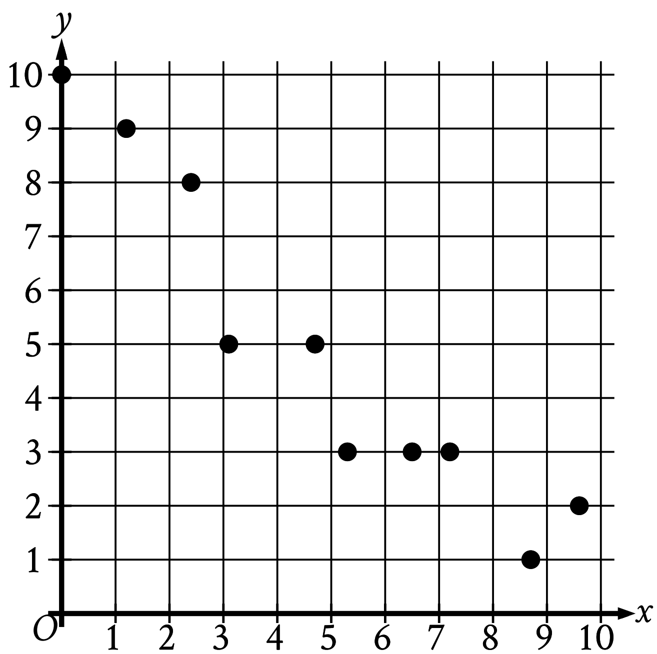
A. 28

B. 40

C. 84

D. 98

The scatterplot shows the relationship between two variables, x and y.



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- The data points have the following approximate coordinates:
  - (0 comma 10)
  - (1.2 comma 9)
  - (2.3 comma 8)
  - (3.1 comma 5)
  - (4.8 comma 5)
  - (5.2 comma 3)
  - (6.5 comma 3)
  - (7.2 comma 3)
  - (9.6 comma 2)
  - (8.8 comma 1)

Which of the following equations is the most appropriate linear model for the data shown?

A. $y = 0.9 + 9.4x$

B. $y = 0.9 - 9.4x$

C. $y = 9.4 + 0.9x$

D. $y = 9.4 - 0.9x$

Q3 PROBLEM-SOLVING AND DATA ANALYSIS

The weights, in pounds, for 15 horses in a stable were reported, and the mean, median, range, and standard deviation for the data were found. The horse with the lowest reported weight was found to actually weigh 10 pounds less than its reported weight. What value remains unchanged if the four values are reported using the corrected weight?

A. Mean

B. Median

C. Range

D. Standard deviation

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

The International Space Station orbits Earth at an average speed of 4.76 miles per second. What is the space station's average speed in miles per hour?

A. 285.6

B. 571.2

C. 856.8

D. 17,136.0

Q5 PROBLEM-SOLVING AND DATA ANALYSIS

To estimate the proportion of a population that has a certain characteristic, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the characteristic is 0.49, with an associated margin of error of 0.04. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the characteristic?

A. It is plausible that the proportion is between 0.45 and 0.53.

B. It is plausible that the proportion is less than 0.45.

C. The proportion is exactly 0.49.

D. It is plausible that the proportion is greater than 0.53.

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

During the first month of sales, a company sold 1,300,000 units of a certain type of smartphone. During the same month, 15% of the units sold were returned. If sales and the return rate remain the same for each of the next 5 months, about how many units of this smartphone will be returned to the company during this 6-month period?

A. 195,000

B. 975,000

C. 1,170,000

D. 6,630,000

United States
Presidents
from 1789 to 2015

Ages	Number
40–44	2
45–49	7
50–54	13
55–59	11
60–64	7
65–69	3

The table above gives the number of United States presidents from 1789 to 2015 whose age at the time they first took office is within the interval listed. Of those presidents who were at least 50 years old when they first took office, what fraction were at least 60 years old?

A. $\frac{10}{43}$

B. $\frac{10}{34}$

C. $\frac{10}{24}$

D. $\frac{25}{34}$

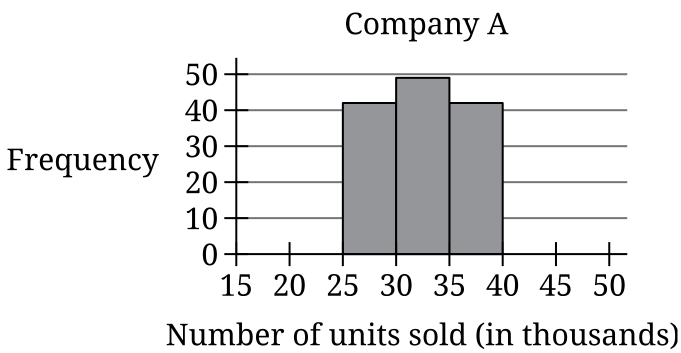
Which of the following is true about the values of 2^x and $2x+2$ for $x>0$?

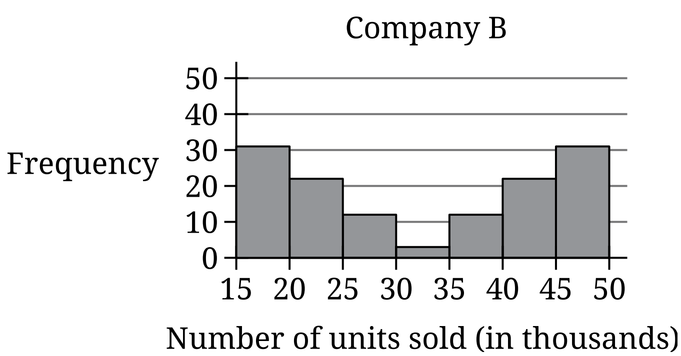
A. For all $x>0$, it is true that $2^x<2x+2$.

B. For all $x>0$, it is true that $2^x>2x+2$.

C. There is a constant c such that if $0<x<c$, then $2^x<2x+2$, but if $x>c$, then $2^x>2x+2$.

D. There is a constant c such that if $0<x<c$, then $2^x>2x+2$, but if $x>c$, then $2^x<2x+2$.





- Company A:
  - The horizontal axis is labeled Number of units sold (in thousands). It ranges from 15 to 50 and is divided into 7 equal intervals.
  - The vertical axis is labeled Frequency. It ranges from 0 to 50.
  - The histogram has 3 bins.
  - The Frequency data for the 3 bins are as follows:
    - 25 to 29: 43
    - 30 to 34: 48
    - 35 to 39: 43
- Company B:
  - The horizontal axis is labeled Number of units sold (in thousands). It ranges from 15 to 50 and is divided into 7 equal intervals.
  - The vertical axis is labeled Frequency. It ranges from 0 to 50.
  - The histogram has 7 bins.
  - The Frequency data for the 7 bins are as follows:

- 15 to 19: 31
- 20 to 24: 24
- 25 to 29: 11
- 30 to 34: 3
- 35 to 39: 11
- 40 to 44: 24
- 45 to 49: 31

The histograms summarize the distributions of number of units sold, in thousands, for company A and company B. Which statement best compares the standard deviations of number of units sold for these companies?

A. The standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.

B. The standard deviation of number of units sold for company A is greater than the standard deviation of number of units sold for company B.

C. The standard deviation of number of units sold for company A is equal to the standard deviation of number of units sold for company B.

D. There is not enough information to compare the standard deviations.

Q10 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

The ratio 140 to m is equivalent to the ratio 4 to 28. What is the value of m?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.